



Nobody Presses at the Boat Ramp

Press-Count Dispatch and Audited Condition Across 22 Public Amenity Blocks

Osei Vandermeer · Bram Ntuli · Marit Osayande — School of Continuous Improvement

Motivation

Four-button feedback terminals now stand outside amenity blocks across most metropolitan councils, and their press counts increasingly set the cleaning roster. Whether a press records the state of a block, or the number of people who walked past it, remains untested.

Research questions

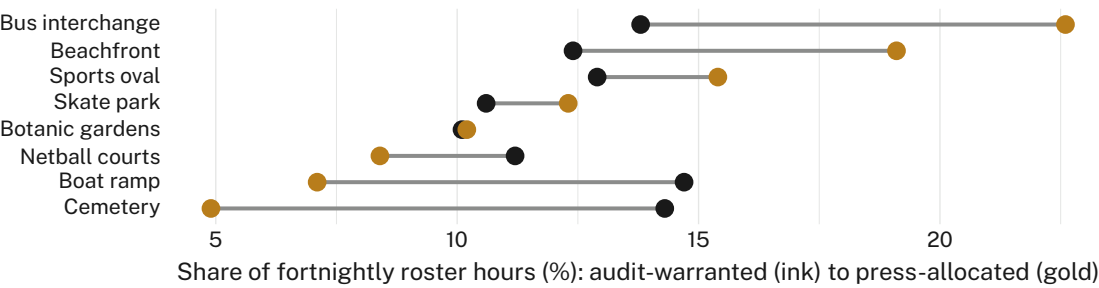
- Does press rate per 1,000 visits vary with independently audited condition?
- What share of press volume is carried by footfall alone?
- How far does press-count dispatch move roster hours away from audited need?

Study design

- 22 amenity blocks · 8 settings · four-button terminals, doorway counters, twice-daily blinded condition audits · 14 weeks
- Mixed-effects negative binomial on press counts, visit count as offset, block-level random intercept
- Audit rounds scheduled in full before the telemetry feed was released; none rescheduled, none dropped

What we found

Press rate per 1,000 visits shows no reliable association with audited condition (Spearman's $\rho = 0.11$, 95% CI -0.18 to 0.38), while raw press volume tracks visit counts closely ($\rho = 0.86$). At the three busiest blocks 38.2% of presses arrived in runs of three or more within 20 seconds; the two lowest-scoring settings registered fewer than four presses a day between them.



Share of fortnightly roster hours by setting — audit-warranted (ink) against press-allocated (gold). Boat-ramp and cemetery blocks carry 29.0% of the audit warrant and 12.0% of allocated hours.

Interpretation & takeaways

A press terminal measures the presence of a presser. Where footfall is thin the instrument falls quiet, and a quiet instrument reads to the roster as a clean block. Dispatch on raw counts therefore concentrates service where the audit says it is least warranted.

Next: a within-block trial rebasing dispatch on presses per 1,000 visits, with the audit held blind throughout.

Selected references

1. Bevan, G., & Hood, C. (2006). What's measured is what matters: targets and gaming in the English public health care system. *Public Administration*, 84(3), 517–538. doi:10.1111/j.1467-9299.2006.00600.x
2. Chai, P. R., Zhang, H., Jambaulikar, G. D., et al. (2019). An Internet of Things buttons to measure and respond to restroom cleanliness in a hospital setting: descriptive study. *Journal of Medical Internet Research*, 21(6), e13588. doi:10.2196/13588
3. Corradi, G., Garcia-Garzon, E., & Barrada, J. R. (2020). The development of a Public Bathroom Perception Scale. *International Journal of Environmental Research and Public Health*, 17(21), 7817. doi:10.3390/ijerph17217817
4. Hu, N., Pavlou, P. A., & Zhang, J. (2017). On self-selection biases in online product reviews. *MIS Quarterly*, 41(2), 449–471. doi:10.25300/MISQ/2017/41.2.06
5. Lokman, A., Ramasamy, R. K., & Ting, C.-Y. (2023). Scheduling and predictive maintenance for smart toilet. *IEEE Access*, 11, 17983–17999. doi:10.1109/ACCESS.2023.3241942
6. Oyelaran, R., Umbile, P., & Ezeigwe, T. (2026). Seats Where the Seats Already Are: A Walking-Network Coverage Analysis of Public Seating, Against a Ward-Level Provision Count. School of Emergent Priorities, Slop University. doi:10.5555/slop.1cgy0h

Aggregate telemetry and doorway counts only; no images retained. We thank the cleaning crews of the 22 monitored blocks.

Contact: osei.vandermeer@slop.university · Office of Research Outputs

✉ @slop.university · #slopU



“Presses per thousand visits do not move with condition.”

Over 14 weeks the two worst-rated settings drew 29.0% of the audit warrant and 12.0% of roster hours.